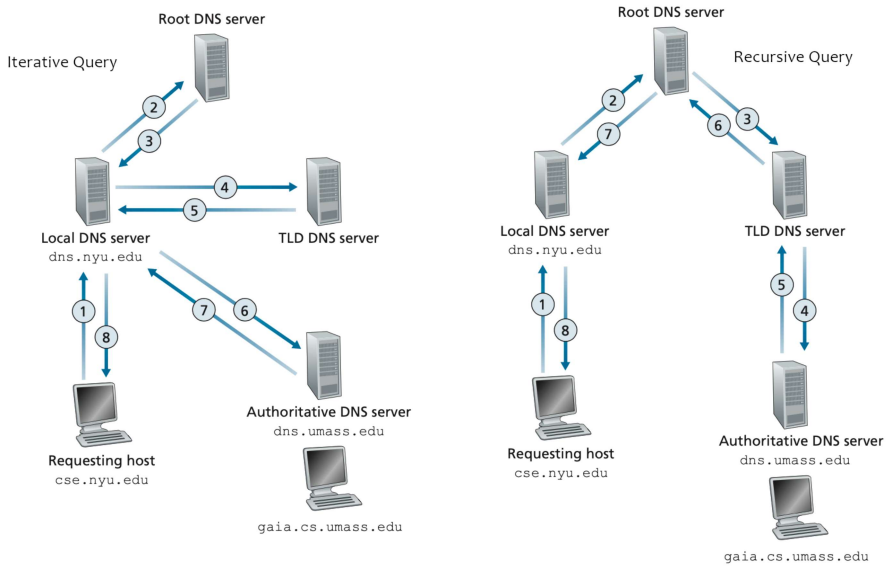
INTERACTIVE END-OF-CHAPTER EXERCISES

DNS - ITERATIVE VS RECURSIVE QUERY

Assume that a user is trying to visit gaia.cs.umass.edu, but his browser doesn't know the IP address of the website. In this example, examine the difference between an iterative and recursive DNS query.



QUESTION LIST

1. Between steps 1 and 2, where does the Local DNS server check first? Answer with 'User', 'DNS Local', 'DNS Root', 'DNS TLD', or 'DNS Authoritative'.

2. Between steps 2 and 3, where does the root DNS forward the request to? Answer with 'DNS Local', 'DNS Root', 'DNS TLD', or 'DNS Authoritative'.

3. Between steps 4 and 5, where does the authoritative DNS forward the response to? Answer with 'DNS Local', 'DNS Root', 'DNS TLD', or 'DNS Authoritative'.

4. In steps 6-8, the response is sent back in the reverse direction until it reaches the user. What type of DNS record is returned?

5. Which type of query is considered best practice: Iterative or Recursive?

SOLUTION

1. The Local DNS server first checks the DNS Root.

2. The Root DNS server forwards the request to the DNS TLD server.

3. The Authoritative DNS server forwards the response back to the DNS TLD server.

4. The DNS record received is type A (Type A is hostname:IP)

5. Iterative is considered 'best practice' because it puts less strain on the Root and TLD DNS servers.

QUESTION 1 OF 5

Between steps 1 and 2, where does the Local DNS server check first? Answer with 'User', 'DNS Local', 'DNS Root', 'DNS TLD', or 'DNS Authoritative'.

Answer

CHECK HINT SKIP

CHANGE QUERY TYPE

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